

Length of a line

Exercise 7.5

Find the length of AB

A (-3, 3) and B(4, -2)

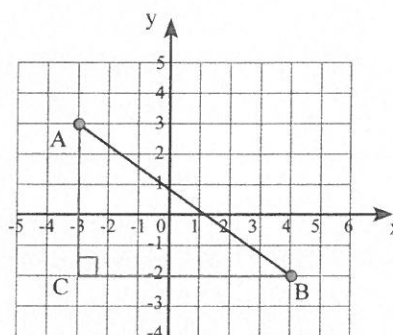
$$AC = 5 \text{ and } CB = 7$$

$$AB^2 = AC^2 + CB^2$$

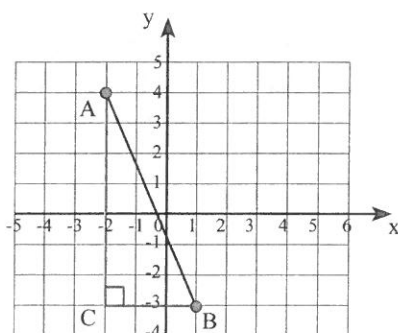
$$= 5^2 + 7^2$$

$$= 74$$

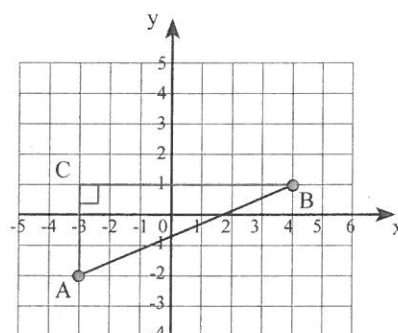
$$AB = \sqrt{74} \text{ or } 8.60$$



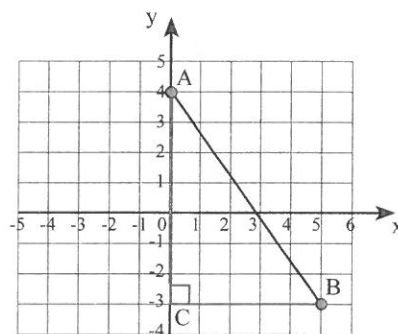
1



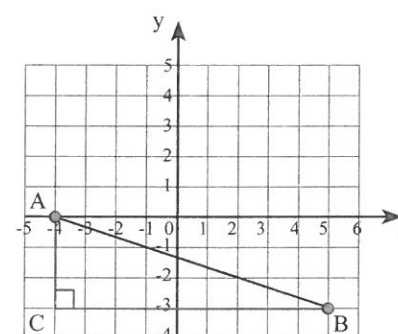
2



3



4



Plot the points first.



Rounding?
See Technology 7.1

5 A(3,2), B(6,5)

6 A(1,1), B(4,5)

7 A(-4,3), B(4,2)

8 A(5,2), B(-5,-3)

9 A(-3,-4), B(5,4)

10 A(-1,5), B(2,-3)

11 A(1,1), B(5,5)

12 A(1,2), B(4,6)

13 A(-4,2), B(4,2)

14 A(5,0), B(0,5)

15 A(-3,3), B(2,3)

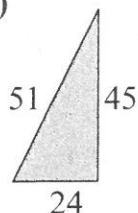
16 A(2,0), B(2,-3)

Chapter Review 1

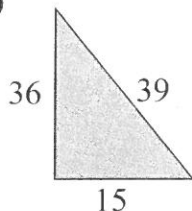
Exercise 7.12

- 1 Which of the following triangles are right-angled triangles?

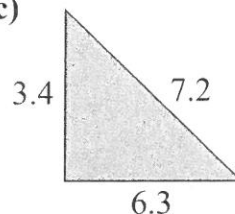
a)



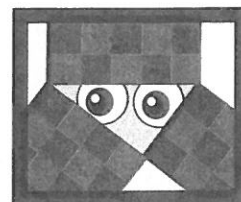
b)



c)



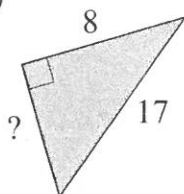
- 2 A rectangular picture frame measures 28 cm by 23 cm with a diagonal of 36 cm. Is the picture frame square?



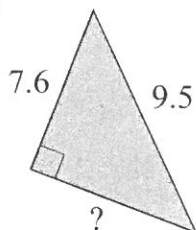
- 3 A carpenter needs to know whether a door frame is square. If the door frame is square then a door can be fitted. The door frame measures 810 mm by 2000 mm and the diagonal is 2158 mm.

- 4 Find the length of the unknown in each of the following:

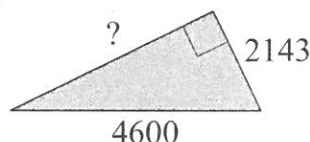
a)



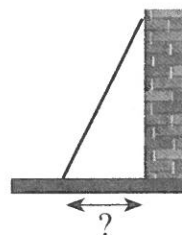
b)



c)

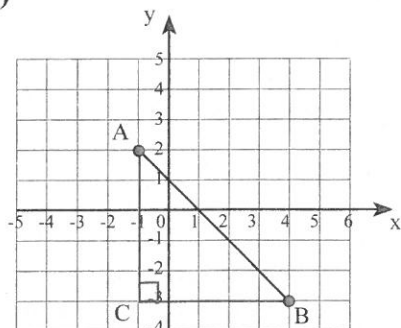


- 5 A 3.2 m ladder is to be laid against a wall so that the top of the ladder is 2.5 m up the wall. How far out from the base of the wall should the ladder be placed?

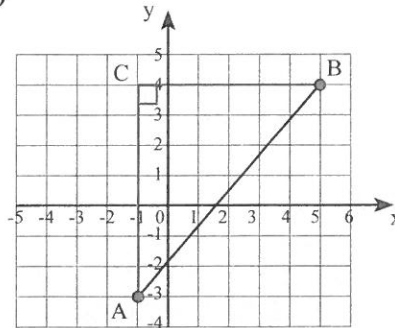


- 6 Find the length of AB.

a)



b)



c) A(5,2), B(6,5)

d) A(1,-1), B(4,4)

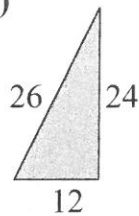
e) A(-4,-3), B(4,3)

Chapter Review 2

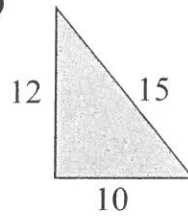
Exercise 7.13

- 1 Which of the following triangles are right-angled triangles?

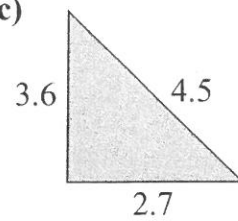
a)



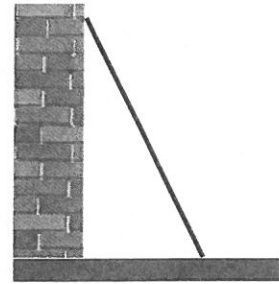
b)



c)

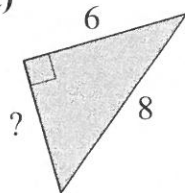


- 2 A 2.2 m ladder is leaning against a wall. The bottom of the ladder is 0.7 m from the wall and the top of the ladder is 2.06 m up the wall. Is the wall vertical?
- 3 A rectangular gate measures 1.4 m by 2.5 m with a 2.79 m diagonal. Is the gate square? If not, should the diagonal be longer or shorter?

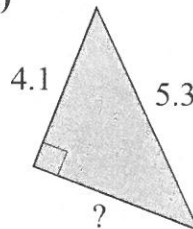


- 4 Find the length of the unknown in each of the following:

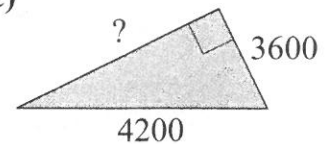
a)



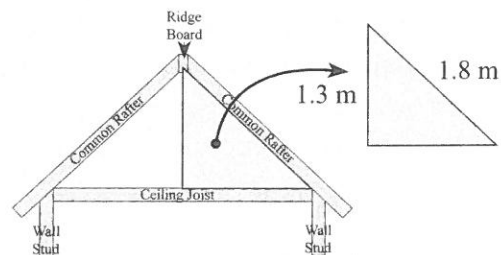
b)



c)

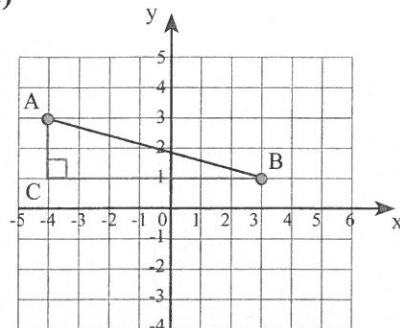


- 5 Calculate the length of the ceiling joist.

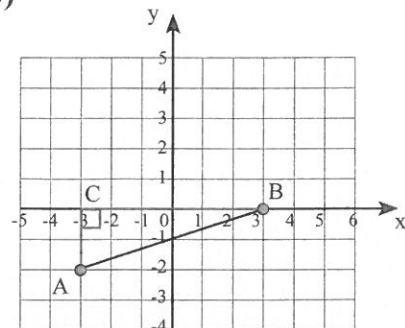


- 6 Find the length of AB.

a)



b)



c) A(3,2), B(6,4)

d) A(-1,1), B(-4,4)

e) A(-4,3), B(3,0)

Exercise 7.2 1 3,4,5-yes 4,5,6-no 6,8,10-yes 5,12,13-yes 7,15,16-no **2a** yes **2b** yes **2c** no
 3 close enough to square **4** no **5** yes **6** no - longer
Exercise 7.3 **1a** 71.59 **1b** 27.86 **1c** 5.09 **2** 3.61 **3** 1445.08 **4a** 68cm **4b** 50cm **4c** 17inch **5** 5.98m **6** 309.23km
Exercise 7.4 **1a** 6.32 **1b** 4.73 **1c** 1313.40 **2** 1.96m **3** 3.65m **4** 24.68m **5** 71.70m
Exercise 7.5 1 7.62 2 7.62 3 8.60 4 9.49 5 4.24 6 5 7 8.06 8 11.18 9 11.31 10 8.54 11 5.66 **12** 5 13 8
 14 7.07 **15** 5 16 3
Exercise 7.6 1 rational-integral **2** irrational **3** rational-decimal **4** rational-decimal **5** rational-decimal
 6 rational-decimal
Exercise 7.7 2 $c^2=a^2+b^2$ **3** yes **4** 10 **5** 1:2 **6** 50mins **7** 40mins **8** 2a **9** $3x+6$ **10** 230
Exercise 7.8 2 $c^2=a^2+b^2$ **3** yes **4** 12 **5** 4:3 **6** 25mins **7** 32mins **8** 2x **9** $5x-10$ **10** 320
Exercise 7.9 2 $c^2=a^2+b^2$ **3** yes **4** 16 **5** 3:2 **6** 36mins **7** 20mins **8** 4x **9** $4x+20$ **10** 180
Exercise 7.10 **1a** 5 **1b** 20 **5a** 7 **5b** $\sqrt{300}$ or $10\sqrt{3}$ **5c** 5 **5d** $\sqrt{85}-\sqrt{45}$ or $\sqrt{85}-3\sqrt{5}$
Exercise 7.11 1 48 **2** $x=18^\circ$
Exercise 7.12 **1a** yes **1b** yes **1c** no **2** close enough to square **3** yes - close enough **4a** 15 **4b** 5.7
4c 4070.33 **5** 2.00m **6a** 7.07 **6b** 9.22 **6c** 3.16 **6d** 5.83 **6e** 10
Exercise 7.13 **1a** no **1b** no **1c** yes **2** close enough to square **3** no **4a** 5.29 **4b** 3.36
4c 2163.33 **5** 2.49m **6a** 7.28 **6b** 6.32 **6c** 3.61 **6d** 4.24 **6e** 7.62